

Geometric Prediction and Exact-Root Certification of Finite-Error Ordering in Quantum Control

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Abstract

Quantum controls that agree at the nominal endpoint and in their complete first-order response to calibrated errors need not have the same finite-error robustness. We study this residual implementation dependence in a serialized two-atom Rydberg model with twenty-four control phases and three global error coordinates.

A fourth-order zero-point response coefficient is first frozen as a prospective ranking rule. On a held-out twenty-path cohort, it achieves Spearman correlation 0.996992 with mean finite-error performance and correctly identifies the best path. The coefficient is then represented as the contraction of a symmetric fourth-order response tensor in the three-dimensional error space with the fourth noise moment; direct recomputation under a nonorthogonal coordinate change verifies the associated covariance and contraction invariance.

For a separate twelve-path cohort, strict 192-bit Krawczyk inclusions certify locally unique roots whose projective output states and first projective responses exactly match the reference. Direct outward-rounded Arb propagation of the certified root boxes through the declared six-axis error functional then proves the complete finite-error ordering for all 66 unordered path pairs, agreeing with an order frozen before the direct propagation. A separate order-thirty local-jet calculation certifies 52/66 comparisons with no reversed certified pair, while the quartic term alone certifies 34/66.

The main result is therefore a computer-assisted exact-root finite-error ordering theorem for the declared finite-dimensional model. It does not constitute PASQAL hardware, model-discrepancy, global-fibre, or many-body evidence.

1 Introduction

Programmable neutral-atom systems expose a rich pulse-level control space, including Rabi amplitude, phase, detuning, atomic geometry, and Rydberg-mediated interactions [1, 2, 3]. This flexibility permits many control schedules to realize the same nominal task. It also creates an identification problem: nominally equivalent implementations can respond differently when calibration errors are finite.

Endpoint equivalence does not resolve this ambiguity. Even matching the complete first-order output-state response to the relevant error coordinates may leave a family of implementations with different higher-order behaviour. The resulting question is not whether leading errors can be cancelled, but whether the remaining finite-error ordering can be inferred before those finite-error outcomes are evaluated.

Composite-pulse and robust-control methods suppress leading error terms [4, 5, 6, 15, 16], and control-landscape theory analyses the critical-point structure of control objectives [13, 14]. Here we instead condition on a strong equivalence relation: candidate controls must share both the nominal output state and the complete first-order output-state response to global amplitude, detuning, and interaction errors. We then ask:

Within the remaining implementation fibre, can a zero-error geometric quantity prospectively predict finite-error robustness, and can that ordering be certified at locally unique projectively matched controls?

This formulation separates two logically distinct tasks. The first is *prediction*: a local quantity must be fixed before held-out finite-error evaluation and must rank previously unseen paths. The second is *certification*: numerical matching must be replaced by exact-root existence and uniqueness, and the finite-error ordering must be enclosed by outward-rounded interval arithmetic, in the validated-numeric tradition of computer-assisted proof [11, 10, 12]. High correlation alone does not provide the second conclusion.

We study a two-atom Rydberg model with twenty-four piecewise-constant phase variables. The numerical matching Jacobian has rank sixteen, leaving an eight-dimensional numerical tangent null space that organizes the candidate implementation fibre.

1.1 Main results

Main theorem (informal statement). Consider the frozen twelve-path cohort, the serialized finite-dimensional two-atom Hamiltonian, the declared transverse phase boxes, and the six-axis mean finite-error functional. For each declared control, a strict Krawczyk inclusion certifies a locally unique root exactly matched in projective output state and first projective response. Direct 192-bit outward-rounded Arb propagation over the twelve certified root boxes produces pairwise disjoint performance intervals whose orientations agree with the pre-outcome frozen ordering for all

$$\binom{12}{2} = 66$$

unordered path pairs. The formal statements and computer-assisted proofs are given in Theorems 1 and 2. This result is conditional on the serialized model and declared error functional; it is not a claim about PASQAL hardware, model discrepancy, global fibre topology, or many-body scaling.

Mechanism proposition (informal statement). Over the same certified Krawczyk boxes, propagation of the zero-point Taylor jet through order thirty, including formal Cauchy-alias and analytic tail enclosures, certifies 52/66 pairwise comparisons with no reversed certified pair. This order-thirty calculation is a separate, computationally distinct mechanism certificate. It is not a premise or computational step in the direct proof of the main theorem. The formal statement appears in Proposition 2.

Reading guide. Four quantitative results play different logical roles:

66/66	direct exact-root theorem
52/66	order-thirty mechanism certificate
34/66	quartic-only boundary
$\rho = 0.996992$	prospective evidence (separate cohort)

The 66/66 direct exact-root certificate is the main theorem. The 52/66 order-thirty result explains how much of that ordering is resolved by a finite local jet. The 34/66 quartic certificate marks the limit of the lowest-order geometric description. The Spearman result is prospective evidence obtained on a separate held-out twenty-path cohort; the frozen order used in the exact-root theorem is supplied by the pre-outcome order-thirty certificate for the twelve-path cohort, and no pathwise identity between the two cohorts is claimed.

The remainder of the paper follows the logical development from prospective prediction to exact-root certification. Section 4 introduces the prospective quartic predictor; Section 5

constructs its covariant tensor form; Section 6 certifies locally unique exact projective matched roots; and Section 7 proves the direct finite-error ordering theorem and reports the separate order-thirty mechanism certificate.

The result is intentionally model-conditional. It concerns the serialized finite-dimensional two-atom Hamiltonian, the declared transverse boxes, and the six-axis mean finite-error functional. It does not establish global fibre topology, worst-case ranking, many-body scaling, open-system robustness, or PASQAL QPU performance.

2 Two-atom control model

2.1 Hamiltonian

We use the ordered basis

$$\{|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle\}.$$

Let

$$\begin{aligned} X_\Sigma &= X \otimes I + I \otimes X, & Y_\Sigma &= Y \otimes I + I \otimes Y, \\ N_\Sigma &= n \otimes I + I \otimes n, & N_{rr} &= n \otimes n, \end{aligned}$$

where $n = |r\rangle\langle r|$. During segment j , the Hamiltonian is

$$\begin{aligned} H_j(\epsilon) &= \frac{\Omega}{2}(1 + \epsilon_\Omega)(\cos \phi_j X_\Sigma + \sin \phi_j Y_\Sigma) \\ &\quad - \Omega \epsilon_\Delta N_\Sigma + V(1 + \epsilon_V)N_{rr}. \end{aligned} \tag{1}$$

The dimensionless error vector is

$$\epsilon = (\epsilon_\Omega, \epsilon_\Delta, \epsilon_V).$$

The nominal parameters are

$$\Omega = 2\pi \text{ rad}/\mu\text{s}, \quad V = 4\Omega,$$

with 24 segments of duration

$$\tau = 0.1 \mu\text{s}, \quad T = 24\tau = 2.4 \mu\text{s}.$$

The nominal interaction can be represented as $V = C_6/R^6$; using the Pulser DigitalAnalogDevice coefficient in the accompanying implementation gives a nominal separation of approximately $7.744 \mu\text{m}$.

For a control path

$$\gamma = (\phi_1, \dots, \phi_{24}) \in \mathbb{T}^{24},$$

the output state is

$$|\psi_\gamma(\epsilon)\rangle = U_{24}(\epsilon) \cdots U_2(\epsilon) U_1(\epsilon) |gg\rangle, \quad U_j(\epsilon) = \exp[-i\tau H_j(\epsilon)], \tag{2}$$

with time-ordered application from segment 1 to segment 24. All primary calculations use exact matrix exponentials for the piecewise constant, finite-dimensional model. Pulser/QutipEmulator was used in an earlier two-atom cross-check, but the formal certificate in this work is independent of Pulser and does not invoke a cloud backend.

2.2 Loss and held-out error law

Let γ_{ref} denote the reference schedule and define

$$|\psi_{\text{ref}}\rangle = |\psi_{\gamma_{\text{ref}}}(\mathbf{0})\rangle.$$

Two infidelities are used. For the extraction of local response coefficients, the target is the path-local nominal state,

$$\mathcal{I}_{\gamma}^{\text{loc}}(\epsilon) = 1 - |\langle \psi_{\gamma}(\mathbf{0}) | \psi_{\gamma}(\epsilon) \rangle|^2. \quad (3)$$

For held-out and direct finite-error comparisons, the target is the common reference state,

$$\mathcal{I}_{\gamma}^{\text{ref}}(\epsilon) = 1 - |\langle \psi_{\text{ref}} | \psi_{\gamma}(\epsilon) \rangle|^2. \quad (4)$$

At an exact response-matched root, $[\psi_{\gamma}(\mathbf{0})] = [\psi_{\text{ref}}]$ projectively, so the two definitions coincide; for numerically matched prospective cohorts, they differ only within the declared matching tolerance. Unless stated otherwise, $\mathcal{I}_{\gamma}$ below denotes $\mathcal{I}_{\gamma}^{\text{ref}}$.

The held-out error design contains the six signed axial errors

$$\mathcal{E} = \{\pm 0.06 e_{\Omega}, \pm 0.04 e_{\Delta}, \pm 0.05 e_V\}. \quad (5)$$

The primary performance functional is the mean

$$\bar{\mathcal{I}}_{\gamma} = \frac{1}{6} \sum_{\epsilon \in \mathcal{E}} \mathcal{I}_{\gamma}(\epsilon). \quad (6)$$

Worst-case infidelity is retained as a secondary diagnostic. The present predictors do not reliably rank the worst case, and no worst-case theorem is claimed.

3 Matched implementation fibre

3.1 Endpoint and first-order constraints

At zero error, remove the unobservable global phase by the path-local horizontal projector

$$P_{\gamma,\perp} = I - |\psi_{\gamma}(\mathbf{0})\rangle\langle\psi_{\gamma}(\mathbf{0})|,$$

and define the path-local horizontal first-order responses

$$J_{\gamma,a} = P_{\gamma,\perp} \left. \frac{\partial}{\partial \epsilon_a} |\psi_{\gamma}(\epsilon)\rangle \right|_{\epsilon=0}, \quad a \in \{\Omega, \Delta, V\}. \quad (7)$$

Since $\psi_{\gamma}(\mathbf{0})$ and ψ_{ref} agree only up to a global phase, align them before comparison by the common phase factor

$$c_{\gamma} = \frac{\overline{\langle \psi_{\text{ref}} | \psi_{\gamma}(\mathbf{0}) \rangle}}{|\langle \psi_{\text{ref}} | \psi_{\gamma}(\mathbf{0}) \rangle|} \in U(1), \quad \tilde{J}_{\gamma,a} = c_{\gamma} J_{\gamma,a}, \quad (8)$$

and compare the phase-aligned response $\tilde{J}_{\gamma,a}$ with $J_{\text{ref},a}$.

Definition 1 (Numerically matched implementation fibre). *For a declared tolerance η , the numerical matched fibre $\mathcal{F}_{\eta}$ consists of controls γ satisfying*

$$\begin{aligned} \mathcal{I}_{\gamma}(\mathbf{0}) &\leq \eta, \\ \|\tilde{J}_{\gamma,a} - J_{\text{ref},a}\| &\leq \eta, \quad a \in \{\Omega, \Delta, V\}, \end{aligned} \quad (9)$$

with $\tilde{J}_{\gamma,a}$ the phase-aligned response of Eq. (8).

3.2 Exact projective constraint map

The dynamics preserves the exchange-symmetric three-dimensional subspace. In a fixed symmetric basis we write the state as

$$\boldsymbol{\psi} = (\psi_0, \psi_1, \psi_2)^T \in \mathbb{C}^3,$$

and eliminate the unobservable global phase by the projective coordinates

$$w_c = \frac{\psi_c}{\psi_0}, \quad c = 1, 2, \quad (10)$$

valid on any region where the chart denominator ψ_0 excludes zero. For each error direction $a \in \{\Omega, \Delta, V\}$, the induced projective first response is

$$\dot{w}_{a,c} = \frac{\dot{\psi}_{a,c} \psi_0 - \psi_c \dot{\psi}_{a,0}}{\psi_0^2}, \quad c = 1, 2, \quad (11)$$

where $\dot{\psi}_a = \partial_{\epsilon_a} \boldsymbol{\psi} \big|_{\epsilon=0}$. The exact constraint map is then

$$F(\boldsymbol{\phi}) = \begin{pmatrix} \Re(w - w_{\text{ref}}) \\ \Im(w - w_{\text{ref}}) \\ \Re(\dot{w}_\Omega - \dot{w}_{\Omega,\text{ref}}) \\ \Im(\dot{w}_\Omega - \dot{w}_{\Omega,\text{ref}}) \\ \Re(\dot{w}_\Delta - \dot{w}_{\Delta,\text{ref}}) \\ \Im(\dot{w}_\Delta - \dot{w}_{\Delta,\text{ref}}) \\ \Re(\dot{w}_V - \dot{w}_{V,\text{ref}}) \\ \Im(\dot{w}_V - \dot{w}_{V,\text{ref}}) \end{pmatrix} \in \mathbb{R}^{16}. \quad (12)$$

The exact matching used below is therefore *exact projective output-state and first-projective-response matching*: it is exact after elimination of the global phase, rather than a component-wise equality of raw state vectors. The numerical optimizer retains the full phase-aligned complex output state and the three full phase-aligned complex state tangents; redundant real components are retained in the residual vector, while singular-value decomposition determines the independent rank.

3.3 Dimension count and generation

The control has 24 phase variables. At the reference solution, the numerical constraint Jacobian has rank

$$\text{rank } DC(\gamma_{\text{ref}}) = 16,$$

leaving an eight-dimensional numerical null space that organizes the candidate implementation fibre. Candidate paths are generated by:

1. drawing a seeded direction in the numerical null space;
2. applying a finite displacement;
3. projecting back to the constraint set by nonlinear least squares;
4. rejecting candidates whose constraint residual or pairwise phase distance fails a predeclared gate.

No held-out finite-error outcome is evaluated during candidate generation. The maximum constraint residual in the reported runs is of order 10^{-11} .

Remark 1 (Global fibre versus local certified roots). *The numerical rank and residual establish a highly accurate computational fibre, suggesting a locally eight-dimensional numerical implementation fibre, not the exact existence of a global eight-dimensional solution manifold. A complete global fibre theorem would require a square transverse formulation and a global interval-Newton or Krawczyk proof. The present work instead establishes 12 locally unique roots exactly matched in projective output state and first projective response via strict Krawczyk inclusions in transverse boxes. The global fibre structure remains open.*

4 L1: Prospective geometric prediction

4.1 Symmetric local response

Introduce normalized error coordinates $z \in \mathbb{R}^3$ through

$$\epsilon = Dz, \quad D = \text{diag}(0.06, 0.04, 0.05).$$

For a direction $v \in \mathbb{R}^3$, define the symmetric response

$$S_\gamma(t, v) = \frac{\mathcal{I}_\gamma^{\text{loc}}(tDv) + \mathcal{I}_\gamma^{\text{loc}}(-tDv)}{2}. \quad (13)$$

Analyticity of the finite matrix exponential gives

$$S_\gamma(t, v) = q_{2,\gamma}(v)t^2 + q_{4,\gamma}(v)t^4 + q_{6,\gamma}(v)t^6 + \dots. \quad (14)$$

Odd powers cancel by construction.

First-order output-state matching fixes the leading quadratic loss to numerical precision. Across the validation cohorts, the relative spread of the averaged quadratic coefficient is approximately 10^{-7} . The first useful discriminator is therefore the quartic coefficient.

4.2 Prospective quartic ranking

The quartic rule developed here is a *geometric predictor*, not a universal theorem: it is gap-dependent and certifies only a subset of pairwise comparisons, a boundary made precise in Sections 5–7.

4.2.1 Discovery and validation separation

The initial four named controls were used only for discovery and orientation. A twelve-path cohort was then used to compare candidate formulas. The smallest sufficient rule on that training cohort was

$$\text{smaller } \mathcal{G}_4 \implies \text{smaller mean finite-error infidelity.}$$

The addition of the sixth-order coefficient did not improve the training rank correlation, so the quartic rule was frozen.

For the L1 prospective audit, the coefficients q_2, q_4, q_6 were estimated by a frozen least-squares fit of

$$S_\gamma(t, e_a) = q_{2,a}t^2 + q_{4,a}t^4 + q_{6,a}t^6$$

at

$$t \in \left\{ \frac{1}{32}, \frac{1}{24}, \frac{1}{20}, \frac{1}{16}, \frac{1}{14}, \frac{1}{12}, \frac{1}{10}, \frac{1}{8} \right\}.$$

The L2 tensor audit below is computationally distinct: its directional derivatives are evaluated directly at $t = 0$ by block-matrix exponentials, rather than by fitting nonzero-radius samples.

A new twenty-path cohort was generated with a different seed. For each candidate, a ranking certificate containing phases, local coefficients, and predicted order was written and SHA-256 hashed before the held-out outcome evaluator was enabled. The predeclared primary gate was

$$\rho_{\text{Spearman}} \geq 0.95,$$

together with correct top-path identification.

4.2.2 Quartic validation result

The twenty-path Colab run gave

$$\rho_{\text{Spearman}}(-\mathcal{G}_4, -\bar{\mathcal{I}}) = 0.996992, \quad p = 1.89 \times 10^{-21}. \quad (15)$$

The predicted and observed top path agreed. Only two adjacent pairs were inverted in the complete ranking. Two local coefficient-extraction windows had rank correlation one, and the maximum local fit residual was 1.48×10^{-10} .

Subsequent cohorts show that $\mathcal{G}_4$ is not universally sufficient. Depending on the generated matched paths, prospective quartic Spearman correlations ranged from approximately 0.85 to 1.00. This variation motivated the bounded higher-order analysis rather than post hoc replacement of the quartic formula.

5 L2: Covariant fourth-order tensor

5.1 Quartic tensor

The object recovered in this section is a quartic susceptibility of the endpoint loss—a component of the fourth jet of the endpoint map composed with infidelity—not a Riemannian curvature. There exists a symmetric fourth-order tensor $A_\gamma^{(4)}$ such that

$$q_{4,\gamma}(v) = A_{\gamma,ijkl}^{(4)} v_i v_j v_k v_l. \quad (16)$$

In three dimensions a symmetric fourth-order tensor has

$$\binom{3+4-1}{4} = 15$$

independent components. The tensor $A_\gamma^{(4)}$ is reconstructed from the fourth jet of the path-local loss $\mathcal{I}_\gamma^{\text{loc}}$: we recover its components from 36 seeded directions, whose directional Taylor coefficients are computed directly at $t = 0$ through order six by block-matrix exponentials, rather than by fitting nonzero-radius samples.

Let

$$z \sim \text{Unif}\{\pm e_\Omega, \pm e_\Delta, \pm e_V\}, \quad \epsilon = Dz,$$

so that the physical error scales $0.06^4, 0.04^4, 0.05^4$ are carried by D and absorbed into the tensor components. For the six-axis law in Eq. (5), define the fourth moment

$$M_{ijkl}^{(4)} = \mathbb{E}[z_i z_j z_k z_l].$$

The scalar quartic predictor is

$$\mathcal{G}_4(\gamma; M^{(4)}) = A_{\gamma,ijkl}^{(4)} M_{ijkl}^{(4)}. \quad (17)$$

For the equally weighted signed axes,

$$\mathcal{G}_4 = \frac{1}{3} \left(A_{\Omega\Omega\Omega\Omega}^{(4)} + A_{\Delta\Delta\Delta\Delta}^{(4)} + A_{VVVV}^{(4)} \right).$$

5.2 Coordinate covariance

Proposition 1 (Covariance and invariant contraction). *Let $z = Ry$ for an invertible real matrix R . Then*

$$A_{y,abcd}^{(4)} = A_{z,ijkl}^{(4)} R_{ia} R_{jb} R_{kc} R_{ld}, \quad (18)$$

$$M_{y,abcd}^{(4)} = (R^{-1})_{ai} (R^{-1})_{bj} (R^{-1})_{ck} (R^{-1})_{dl} M_{z,ijkl}^{(4)}, \quad (19)$$

and

$$A_y^{(4)} : M_y^{(4)} = A_z^{(4)} : M_z^{(4)}. \quad (20)$$

Proof. Equation (18) follows by substituting $z = Ry$ into the quartic form. Equation (19) follows from $y = R^{-1}z$. Contracting all four indices cancels each R with the corresponding R^{-1} , proving Eq. (20). $\square$

This proposition is algebraic. The numerical audit additionally recomputes the tensor directly in the transformed coordinates. For a nonorthogonal map of condition number approximately 1.82, the direct-refit relative error is 1.08×10^{-14} and the maximum invariant-contraction relative error is 2.36×10^{-15} .

The covariant object is not the scalar $\mathcal{G}_4$ in isolation. It is the fourth-order response tensor $A^{(4)}$ together with the physical fourth noise moment $M^{(4)}$. Their contraction is independent of a linear reparameterization of error coordinates. This avoids interpreting a coordinate-specific axis average as an intrinsic curvature.

Establishing a relation of this susceptibility to a canonical connection or curvature on a control quotient would require additional structure.

For a noise moment M , the low-cost screening problem is

$$\gamma^* \in \arg \min_{\gamma \in \mathcal{F}_\eta} A_\gamma^{(4)} : M^{(4)}.$$

Pairs that fail the quartic gap test can be escalated adaptively to sixth, eighth, or higher jet order until their intervals separate. This suggests a certified branch-and-bound control selector: compute low-order local tensors for all candidates, refine only ambiguous pairs, and reserve direct finite-error simulation or hardware evaluation for model validation rather than exhaustive ranking.

The gap-dependent certification criterion used by the implementation retains the residual constant and quadratic terms explicitly. The certified quantity is the six-axis averaged reference loss at radius t ,

$$\bar{\mathcal{I}}_k^{\text{ref}}(t) = \frac{1}{6} \sum_{a \in \{\Omega, \Delta, V\}} \sum_{\sigma = \pm 1} \mathcal{I}_{\gamma_k}^{\text{ref}}(\sigma t D e_a). \quad (21)$$

For each path k write the truncated local model and its enclosure as

$$P_k(t) = C_{0,k} + C_{2,k} t^2 + \mathcal{G}_{4,k} t^4, \quad \bar{\mathcal{I}}_k^{\text{ref}}(t) \in P_k(t) + [-B_k(t), B_k(t)]. \quad (22)$$

If

$$|P_a(t) - P_b(t)| > B_a(t) + B_b(t), \quad (23)$$

then the sign of $\bar{\mathcal{I}}_a^{\text{ref}}(t) - \bar{\mathcal{I}}_b^{\text{ref}}(t)$ is strictly certified. Only when $C_{0,a} = C_{0,b}$ and $C_{2,a} = C_{2,b}$ does this reduce to the simplified quartic-gap form

$$|\mathcal{G}_{4,a} - \mathcal{G}_{4,b}| |t|^4 > B_a(t) + B_b(t).$$

The implementation folds the residual C_0 and C_2 differences into the correction intervals, so they are not omitted from the certificate.

6 L3: Certified exact roots via Krawczyk inclusions

The numerical paths satisfy the endpoint and first-order matching constraints to order 10^{-11} . To move from numerical candidates to exact mathematical roots, we use the projective constraint map $F : \mathbb{R}^{24} \rightarrow \mathbb{R}^{16}$ of Eq. (12) and decompose the phase space into 16 transverse variables x and 8 fibre variables y . For each declared numerical path γ_k , we fix its fibre coordinates y_k and consider the reduced square map

$$F_k(x) = F(x, y_k).$$

For a box $X \subset \mathbb{R}^{16}$ with centre x_0 , the Krawczyk operator is

$$K(X) = x_0 - CF(x_0) + (I - C[DF(X)])(X - x_0), \quad (24)$$

where C is a frozen nonsingular numerical preconditioner (an approximate inverse of the midpoint Jacobian) and $[DF(X)]$ is an interval enclosure of the Jacobian over X . In centered coordinates $x_0 = 0$, as used by the implementation, this reduces to

$$K(X) = -CF(0) + (I - C[DF(X)])X.$$

The strict inclusion $K(X) \subset \text{int}(X)$ then implies that F_k has a unique zero inside X [17, 9].

The projective chart denominator ψ_0 of Eq. (10) is formally verified to exclude zero on every accepted box, ensuring the coordinate chart is valid throughout the Krawczyk computation.

Theorem 1 (Krawczyk inclusions for 12 declared controls). *For each of the twelve declared controls, there exists a transverse phase box*

$$X_k \subset \mathbb{R}^{16}$$

such that the Krawczyk operator satisfies

$$K_k(X_k) \subset \text{int}(X_k).$$

Consequently, each box X_k contains a locally unique exact root x_k^ of $F_k(x) = 0$, i.e., a control $\gamma_k^* = (x_k^*, y_k)$ that is exactly matched in projective output state and first projective response to the reference.*

Computer-assisted proof. A numerical approximate inverse of the midpoint Jacobian is used as the point preconditioner. The Jacobian over each full box is then enclosed with 192-bit Arb arithmetic, and the resulting Krawczyk image is verified to lie strictly inside the box. The projective chart denominator is formally verified to exclude zero on every accepted box. For all twelve boxes, the inclusion $K(X) \subset \text{int}(X)$ is verified. The protocol hash is

a19d01ec77a783be4e3858dd92ca080a
a9fde2b09b77541b0e1ad7f392ff3124

and the certificate hash is

fc3ad3823597bd5281b7b9292f227f72
3c2b3a9a8aa4e42c9f0353921b4d8fec.

By the Krawczyk theorem [17, 9, 10], each box contains a locally unique root. $\square$

Remark 2 (Scope of L3). *The theorem establishes local uniqueness inside each declared transverse box. It does not certify global uniqueness on the entire fibre, nor does it prove that every point on the numerical fibre has an exact representative. The global fibre structure remains open.*

7 L4: Exact-root finite-error ordering

7.1 Direct Arb propagation of Krawczyk boxes

The following theorem is the main result of this work. Its proof uses only the certified Krawczyk boxes and direct outward-rounded propagation of the finite-error functional. No Taylor truncation, Cauchy extraction, alias enclosure, or local-jet reconstruction is used.

Theorem 2 (Exact-root finite-error ordering). *For each of the twelve declared controls, let X_k be the transverse phase box satisfying a strict Krawczyk inclusion from Theorem 1. Direct outward-rounded Arb propagation of X_k through the declared six-point finite-error functional produces twelve pairwise ordered performance intervals. All 66 unordered path pairs are disjoint and agree with the prospectively frozen ordering. Therefore the locally unique roots exactly matched in projective output state and first projective response inside the boxes obey the complete frozen finite-error ordering.*

Computer-assisted proof. By Theorem 1, the locally unique projectively matched control γ_k^* belongs to the certified phase box X_k .

For every box X_k , the six declared finite-error Hamiltonians are evaluated directly using 192-bit outward-rounded Arb/ACB arithmetic. Every phase-dependent trigonometric factor, Hamiltonian matrix exponential, state propagation, overlap, infidelity, and six-point mean is enclosed. Denote the resulting real performance interval by B_k . Therefore

$$\bar{\mathcal{I}}(\gamma_k^*) \in B_k.$$

For every frozen ordered pair $a \prec b$, direct interval comparison verifies

$$\sup B_a < \inf B_b.$$

All $\binom{12}{2} = 66$ comparisons pass and have the orientation specified by the frozen ordering certificate. Hence

$$\bar{\mathcal{I}}(\gamma_a^*) < \bar{\mathcal{I}}(\gamma_b^*)$$

for every frozen ordered pair. No Taylor truncation or local-jet reconstruction is used in this direct certification.

The predicted ordering was fixed by the predecessor zero-point ordering certificate before the exact-root direct finite-error propagation was evaluated. The final exact-root certificate, containing the formal direct intervals and all pairwise comparisons, was written and hashed after the propagation.

The exact-root ordering protocol hash is

```
f51ecfd2e4572f271e276cae18697005
baa3f0ec3f3ab94f98f919cd50c3f844
```

and the certificate hash is

```
03c4db75cb80a149ce1a61e11c227039
f9bd79c3f1304b3defa311a491ea2d1e.
```

□

Remark 3 (Scope of Theorem 2). *The theorem is conditional on the serialized-decimal finite Hamiltonian model, the declared transverse boxes, and the six-axis mean error functional. It does not certify model discrepancy, global fibre topology, many-body scaling, or PASQAL hardware. It also does not prove a global eight-dimensional exact fibre theorem; it proves finite-error ordering for the locally unique exact roots inside the declared boxes.*

Remark 4 (Integrity check). *In all reported runs, every ordinary held-out floating-point outcome is verified to lie inside its corresponding formal ball. This serves as an integrity diagnostic and is not part of the theorem proof.*

7.2 Order-thirty mechanism certificate

The following proposition provides a separate, computationally distinct mechanism certificate using the zero-point jet. Unlike the main theorem, it uses Taylor coefficients, Cauchy extraction, alias enclosures, and Dyson tail bounds. It is not part of the main theorem proof.

The floating-point precursor obtains the order-thirty jet by block-matrix exponentials. The formal phase-box certificate used in Proposition 2 instead extracts the coefficients with a 64-point outward-rounded Cauchy transform.

Proposition 2 (Order-thirty mechanism certificate). *Propagation of the zero-point jet through order thirty over the twelve Krawczyk phase boxes, including the formal Cauchy-alias and analytic order-32 tail enclosures, certifies 52/66 frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified.*

Computer-assisted proof. For path k and physical error axis a , define the analytic transition amplitude

$$g_{k,a}(\lambda) = \langle \psi_{\text{ref}} | \psi_{\gamma_k}(\lambda D e_a) \rangle = \sum_{n \geq 0} g_{k,a,n} \lambda^n.$$

Its coefficients are extracted by a 64-point outward-rounded Cauchy discrete Fourier transform on the complex unit circle:

$$\hat{g}_n = \frac{1}{64} \sum_{k=0}^{63} g(\zeta_k) \zeta_k^{-n}, \quad \zeta_k = e^{2\pi i k / 64}. \quad (25)$$

The discrete transform aliases coefficients $g_{n+64\ell}$. A path-uniform Dyson bound encloses this alias. The implemented enclosure is 10^{-40} , and the formal gate verifies that

$$e^{K_a} \frac{K_a^{64}}{64!} < 10^{-40}$$

for every physical error axis, where the dependence is parameterized as

$$H(s; \lambda_a) = H_0(s) + \lambda_a H_{1,a}(s),$$

with s the physical time and λ_a the dimensionless error parameter along axis a , and

$$K_a = \int_0^T \|H_{1,a}(s)\|_2 \, ds = \int_0^T \left\| \frac{\partial H(s; \lambda_a)}{\partial \lambda_a} \Big|_{\lambda_a=0} \right\|_2 \, ds$$

is the integrated spectral norm of the Hamiltonian perturbation along axis a .

Fidelity coefficients are formed by ball convolution,

$$F_n = \sum_{r=0}^n \overline{g_r} g_{n-r},$$

and the infidelity coefficients follow from $1 - F$. The analytic even Dyson tail after order thirty is enclosed by 2×10^{-11} .

Propagating the resulting order-thirty jet over the twelve Krawczyk phase boxes yields performance intervals. Pairwise comparison of the resulting order-thirty intervals certifies 52/66 frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified. The unresolved pairs arise from dependency and wrapping in the phase-box propagation of the jet enclosure. They are not reversed physical comparisons and do not indicate an omitted-tail failure. $\square$

Table 1: Representative prospective and formal audits. “Pairs” denotes certified unordered comparisons. Candidate sets differ between stages and between numerical environments; every stage freezes its protocol and ranking before held-out evaluation. The direct exact-root ordering is the main theorem.

Audit	Paths	Spearman ρ	Certified pairs	Status
Prospective G_4 prediction	20	0.996992	–	pass
Covariant fourth-order tensor	12	0.965035	–	pass
Formal quartic-only certificate	12	–	34/66	partial
Frozen-point order-30 certificate	12	1.000000	66/66	pass
Exact-root Krawczyk inclusion	12	–	12/12	pass
Exact-root order-30 mechanism	12	–	52/66	partial
Exact-root direct Arb ordering	12	1.000000	66/66	pass

Table 2: Numerical integrity diagnostics from representative Colab runs.

Diagnostic	Value
Fourth-tensor independent components	15
Maximum tensor recovery relative residual	4.63×10^{-15}
Direct coordinate-refit relative error	1.08×10^{-14}
Invariant-contraction relative error	2.36×10^{-15}
Quadratic-coefficient relative spread	1.09×10^{-7}
Arb precision	192 bits
Cauchy points	64
Taylor order	30
Formal quartic pair coverage	34/66 = 51.52%
Exact-root order-30 mechanism coverage	52/66 = 78.79%
Exact-root direct Arb ordering coverage	66/66 = 100%

8 Results

The sequence of results supports the following hierarchy.

1. **Implementation memory:** endpoint and complete first-order output-state matching do not numerically determine finite-error performance.
2. **Low-order organization:** the covariant quartic contraction $\mathcal{G}_4$ strongly organizes mean performance and formally certifies a substantial subset of path pairs (34/66 at the formal ball level).
3. **Exact-root certification:** strict Krawczyk inclusions establish 12 locally unique roots exactly matched in projective output state and first projective response.
4. **High-order mechanism:** the order-thirty jet propagated over the Krawczyk boxes certifies 52/66 pairs with zero reversals (Proposition 2).
5. **Complete ordering:** direct 192-bit outward-rounded Arb propagation of the Krawczyk boxes certifies all 66 pairs with the frozen order preserved (Theorem 2).

9 Discussion

9.1 Why the quartic term is useful but incomplete

First-order matching makes the quadratic loss nearly common. This promotes the quartic term to the first visible discriminator. It explains why $\mathcal{G}_4$ often gives almost perfect rankings. However, if two quartic coefficients are close, the sixth and higher terms can reverse their order. The formal intervals expose this distinction directly: 34 pairs are quartic-certified, while the remaining pairs require higher jet data or direct propagation.

This is preferable to declaring the quartic rule either universally true or false. The correct statement is gap-dependent as in Eq. (23).

9.2 Order-30 mechanism versus direct propagation

The order-30 jet propagated over the Krawczyk boxes certifies 52/66 pairs with zero reversals (Proposition 2). The remaining 14 pairs are not physically reversed—they are unresolved due to dependency and wrapping in the phase-box propagation of the jet enclosure. Direct Arb propagation of the same boxes certifies all 66 pairs (Theorem 2), demonstrating that the unresolved cases are an artefact of the jet propagation and interval wrapping, not a physical failure of the frozen ordering. The 52/66 mechanism certificate and the 66/66 direct certificate arise from different propagation levels and should not be conflated: the latter is the main theorem.

The content is quantitative rather than existential: an analytic finite-dimensional propagator is determined by its local germ, but here a fixed finite order, 30, resolves 52/66 comparisons with an explicitly bounded tail, and the unresolved cases arise from interval wrapping rather than truncation.

10 Limitations and next theorem

The present work has five explicit limitations.

1. **Global fibre structure.** Twelve locally unique roots exactly matched in projective output state and first projective response are certified in transverse boxes. The global eight-dimensional solution manifold structure remains unproved.
2. **Finite-dimensional model.** Spontaneous emission, Doppler effects, laser phase noise, position fluctuations, leakage levels, and waveform filtering are absent.
3. **Two atoms.** No many-body scaling theorem is claimed.
4. **Mean axial error law.** The primary result concerns the declared six-axis mean. Worst-case ranking is not supported by the present scalar.
5. **No QPU evidence.** The calculations are local and require no PASQAL credentials. They do not constitute PASQAL Cloud, FRESNEL, or physical-QPU evidence.

In short, the certificate is formal for the serialized finite-dimensional two-atom Hamiltonian, the declared transverse boxes, and the six-axis mean error functional; it does not certify model discrepancy, global fibre topology, many-body scaling, or PASQAL hardware.

The next mathematical step is a validated implicit-function theorem or a parameter-dependent Krawczyk atlas: treat the eight fibre coordinates as parameters and certify a unique transverse solution varying over a declared fibre neighbourhood. The target mathematical structure is

$$F(x, y) = 0, \quad x \in \mathbb{R}^{16}, \quad y \in \mathbb{R}^8,$$

and the goal is to prove that a unique transverse solution $x = x(y)$ exists and is regular over a declared y -region, not to prove a single global isolated root.

11 Reproducibility

The principal standalone scripts are:

```
pasqal_two_atom_G4_standalone_colab.py
pasqal_L3_L4_standalone_colab.py
pasqal_L4_order30_standalone_colab.py
pasqal_L4_formal_arb_standalone_colab.py
pasqal_L4_exact_fibre_krawczyk_standalone_colab_v1_2.py
pasqal_L4_exact_root_ordering_standalone_colab.py
pasqal_L4_reproducible_certificate_v1_3_colab.py
```

The last script produces all formal proof objects.

The v1.3 standalone freezes the twelve-path cohort, transverse bases, and point preconditioners, fixes the common Krawczyk radius at 3×10^{-12} , and excludes runtime metadata from every hashed proof object. Two complete formal executions produce byte-identical protocol, certificate, and report files. The frozen cohort hash is

```
9942af610c1e9499a10e989e4ae069e3
912d58eec4fd3ad1d3d39947c232c356.
```

The deterministic protocol and certificate hashes are:

- Krawczyk protocol:

```
a19d01ec77a783be4e3858dd92ca080a
a9fde2b09b77541b0e1ad7f392ff3124
```

- Krawczyk certificate:

```
fc3ad3823597bd5281b7b9292f227f72
3c2b3a9a8aa4e42c9f0353921b4d8fec
```

- Exact-root ordering protocol:

```
f51ecfd2e4572f271e276cae18697005
baa3f0ec3f3ab94f98f919cd50c3f844
```

- Exact-root ordering certificate:

```
03c4db75cb80a149ce1a61e11c227039
f9bd79c3f1304b3defa311a491ea2d1e
```

The formal standalone script pins `python-flint==0.8.0`. NumPy and SciPy provide numerical diagnostics, while the v1.3 proof consumes the frozen decimal cohort and uses Arb/ACB for the formal interval certificate.

Clean-environment reproduction. The complete v1.3 audit was rerun in a clean external Colab environment from the public v0.3.1 Git tag. The tag resolved to commit 284974c9f6b952f4e114c8c5bdc9c2c299c4065c. Dependency installation, repository SHA-256 verification, reference-artifact validation, regression tests, and the complete two-run formal recomputation all passed. The run required no PASQAL account or cloud credentials.

12 Conclusion

We establish two complementary results on distinct frozen cohorts. On a prospective twenty-path cohort, the zero-point quartic geometric scalar strongly predicts held-out finite-error performance. On a separate twelve-path formal cohort, direct outward-rounded interval propagation certifies all 66 pairwise finite-error orderings at locally unique roots exactly matched in projective output state and first projective response.

The common endpoint and complete first-order state response leave residual path dependence. A coordinate-covariant fourth-order response tensor provides a strong and partially certifying low-order description, but close comparisons require higher derivatives. On the twelve-path cohort, strict Krawczyk inclusions establish the twelve locally unique exact roots. The zero-error jet through order thirty, combined with rigorous Cauchy-alias and Dyson-tail enclosures, certifies 52/66 exact-root comparisons with zero reversals as a separate mechanism certificate. Direct 192-bit outward-rounded Arb propagation of the Krawczyk boxes certifies all 66 pairwise orderings as the main theorem.

Establishing a global fibre theorem and extending the certificate to open-system, many-atom, and hardware data are the remaining steps.

Code and Data Availability

All code, frozen inputs, formal certificates, and reproduction scripts are available at

<https://github.com/papasop/quantum-control-geometry>

The formal computational artifacts used in this work are fixed in release `v0.3.1` at commit `284974c9f6b952f4e114c8c5bdc9c2c299c4065c`. The synchronized manuscript source is maintained on the repository’s `main` branch. No PASQAL account or cloud credentials are required.

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The author declares no competing interests.

Author Contributions

Y.Y.N. Li conceived the study, developed the theoretical framework, implemented the numerical and formal audits, analyzed the results, and wrote the manuscript.

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